

Li₃PS₄ | Amorphous Sulfide

Transferability of pretrained-potential structure and transport beyond oxychlorides

1 Literature

Mirmira et al. (2021): experiment
Chen et al. (2025): DeePMD
Reference T: 300–900 K

2 Model & preparation

Li₁₉₂P₆₄S₂₅₆; 512 atoms
NEP89 / GPUMD
1500 K NPT: 100 ps
Cool 2.5 K ps⁻¹ → 300 K
300 K hold: 20 ps

3 Transport MD

T: 300 / 500 / 700 / 900 K
Ramp: 10 ps
NPT (1 bar): 50 ps
NVT production: 200 ps
dt: 0.5 fs

4 Analysis

Li MSD → D(T)
Li-S / P-S RDF and CN
S-P-S angle
Framework motion

5 Conclusion

Local motifs partly retained
Transport magnitude differs
Dedicated work remains